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Building a Lightweight Digital Twin of a Crane Boom for Structural Safety Monitoring Based on a Multifidelity Surrogate Model

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Undetected fatigue and overload damages at the key locations of the crane boom are among the biggest threats in construction, leading to structural failure. Thus, the structural health of the crane boom should be monitored in real time to ensure that it works under the designed load capacity. In this study, we developed a lightweight digital twin by the multifidelity surrogate (MFS) model to improve the real-time monitoring and prediction accuracy of the structural safety of a crane boom. Digital twin technology, which can establish real-time mapping between the physical space and the digital space, has a promising potential for online monitoring and analysis of structures, equipment, and even human bodies. By combining the MFS model and sensor data, the lightweight digital twin can dynamically mirror the crane boom postures and predict its structural performance in real time. In this study, the structural analysis of the crane boom is limited to the linear elastic stage of materials. Numerical experiments showed that the accuracy of the lightweight digital twin was enhanced compared with that established by the single-fidelity surrogate model, and the computational cost of the lightweight digital twin was decreased with respect to the digital twin built by the numerical method. Meanwhile, the uncertainty from the physical space was analyzed to enhance the reliability of the lightweight digital twin. Thus, the lightweight digital twin developed in our work can ensure accurate safety prediction and design optimization for crane booms. [DOI: 10.1115/1.4053606]

Keywords: digital twin, safety monitoring, multifidelity surrogate model, crane boom, design automation

1 Introduction

Cranes are one of the most widely used heavy equipment in factories, construction sites, and ports. As a key component, the crane boom deserves great attention in terms of structural safety as accidents caused by the structural damage of the crane boom may cause disastrous consequences [1,2]. By monitoring, analyzing, and predicting the structural status of the crane boom in real time, the occurrence of such accidents can be decreased to a certain extent. Therefore, an effective way is required to quickly and reliably predict the structural performance of any key location on the crane boom and improve its design. As a promising solution, the digital twin is widely used for online monitoring, analysis, and prediction of various industrial devices including aircraft, vessels, and automobiles. Therefore, the digital twin was adopted in this study for the safety monitoring and prediction of the crane boom.

As initially defined by Grieves [3], the digital twin contains physical products, virtual products, and connections between them. Since its introduction, the concept, its implementation, and applications have been heavily studied [4–8]. NASA and the US Air Force have applied digital twin technology to the condition monitoring, fault diagnosis, life prediction, and design optimization of various types of air vehicles by integrating diverse heterogeneous information sources [9–11]. In addition, numerous studies have demonstrated its applications in diverse equipment, including mobile cranes [12], wind turbines [13], and computer numerical control machine tools [14]. The digital twin has transformed from a descriptive concept to an actionable technique.

In particular, structural safety monitoring is one of the most promising applications of the digital twin as it can obtain the real-time reflection of the physical world and facilitate decision-making based on fast analytical algorithms. By taking the real forces or displacements obtained from sensors as inputs, Haag and Andrel developed a digital twin that employed the finite element method (FEM) to perform the structural analysis of a simple bending beam [15]. Guivarch et al. proposed an approach to building a digital twin for a swashplate rotor assembly of a helicopter using multibody simulations [16]. Fotland et al. performed multiple simulation methods and efficiently built a digital twin for a crane pulley and cable [17]. Moi et al. implemented a digital twin for the condition monitoring of a knuckle boom crane based on an inverse

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method using strain gauges as load sensors, which provided real-time determinations of stresses, strains, and loads at an unlimited number of spots [18]. These studies focused on combining mechanistic models, such as structural statics and dynamics with numerical methods to establish digital twins. The essential benefit of using this way to build the digital twin is that the model parameters have an actual physical meaning, which facilitates the scientific interpretation of the results.

However, numerical methods are typically difficult to be used in real-time applications due to their high computational consumption. To overcome this limitation, Rasheed et al. suggested using artificial intelligence (AI) and machine learning (ML) to build the digital twin [19]. Von Rueden et al. described a hybrid modeling approach by combining AI/ML algorithms and simulations, aimed at bridging the gap between data-based and knowledge-based modeling [20]. Magargle et al. introduced a simulation-based digital twin to support heat monitoring and predictive maintenance for an automotive braking system [21]. Wang et al. presented a digital twin for rotating machinery fault diagnosis based on rotor dynamics and the FEM [22]. Li et al. employed a dynamic Bayesian network to build a digital twin for monitoring the crack growth on the leading edge of an aircraft wing [23]. Essentially, the currently popular AI and ML methods utilize statistics to predict trends and patterns. To use digital twins in structural safety monitoring, a common way to introduce AI/ML is to use the results of numerical analysis as the training data to build the AI/ML model. In this way, AI/ML model equations are derived by developing a probabilistic model that best describes the relationship between dependent and independent variables. However, since the statistical model is based on correlation rather than causality, it cannot provide an interpretation of the mechanistic process.

Both mechanistic and statistical models have their advantages and shortcomings. Therefore, a more reasonable way to combine simulation with AI/ML should be adopted to build the digital twin for the structural analysis. One promising approach was presented by Willcox and her coworkers by combining a library of component-based reduced-order models with Bayesian state estimation or optimal tree algorithm to create data-driven physics-based digital twins [24,25]. Compared with an optimal tree, the Bayesian state estimation could quantify the uncertainty of the digital twin. Furthermore, they proposed a probabilistic graphical model as a formal mathematical representation of the digital twin and its associated physical asset, which helped to move the digital twin from a one-off model to an accessible robust model at scale [26]. Another way to combine simulations with AI/ML was realized simply by training the simulation results based on several data points of samples, while other data points in the solution space were interpolated or fitted [27]. However, a reliable model built by this way generally requires substantial samples, which will consume considerable computational resources. The two combination-based approaches are limited by high computational costs. Therefore, if the computational effort is reduced and the prediction accuracy is improved, the digital twin composed of mechanistic and statistical models will be more lightweight, efficient, and reliable.

The multifidelity surrogate (MFS) model widely used in engineering design and optimization is adopted to achieve this goal. A high-fidelity (HF) model is generally accurate but also more time consuming than the corresponding low-fidelity (LF) model [28]. A general trend of the outputs of interest can be provided by a large number of inexpensive LF samples, and the accuracy of outputs can be improved by a few expensive HF samples [29]. Thus, the MFS model can not only greatly reduce the computational effort of the numerical methods but also improve the accuracy of the single-fidelity surrogate model. In this study, a lightweight digital twin for structural safety monitoring based on the MFS model was proposed by adopting both statistical and mechanics-based models, and the data obtained from the models and multiple sensors were combined to provide more accurate results and faster solutions. Notably, the simulation of the crane boom was limited to the linear elastic stage of materials. The MFS model-based lightweight digital twin focuses on the key locations of equipment, is more accurate, efficient, and lightweight, and uses fewer computing resources to achieve higher accuracy compared with the digital twin proposed in our previous work [27].

The remainder of this article is organized as follows. In Sec. 2, the method used to establish the MFS model-based lightweight digital twin for structural safety monitoring is introduced. Section 3 presents the implementation process of the digital twin in detail by taking a crane boom as an example. Compared with the results obtained from the HF computer simulation, the effectiveness of the proposed digital twin is verified in Sec. 4, as well as the uncertainty analysis of the lightweight digital twin is performed. Discussion and conclusions are presented in Sec. 5.

2 Composition of the Lightweight Digital Twin

2.1 Computer Simulation Model. When taking into account the structural performance information, the current digital twin is generally expressed by various mechanistic models and calculated by multiple numerical methods, including the FEM, finite volume method, finite difference method, etc. In this study, the FEM was adopted to execute the simulation for the structural performance of the crane boom. However, running HF finite element analysis might be extremely time consuming, especially for systems with high-dimensional inputs. Therefore, it is significant to conduct the discretization of input variables and the simplification of the system for building a digital twin. The input variables of engineering problems mainly contain shape, posture, velocity, load, and so on. The combination of these input variables is different in various states of the same system, and it is difficult to simulate a system that contains all these different states simultaneously. Discretization can significantly simplify this problem and break the limitation. When simulating a system, the input variables are divided into several dimensions, and each dimension is further discretized based on which the entire solution space can be represented through sampling. In this study, two types of computer simulations are executed: the LF finite element (LFFE) model and the HF finite element (HFFE) model. In this study, the LFFE model is built by three-dimensional (3D) beam element, while the HFFE model is built by the tetrahedral element. The key to building the digital twin is to take advantage of both the LFFE and HFFE models to improve computational efficiency while meeting certain accuracy requirements.

2.2 Multifidelity Surrogate Model. Surrogate models can be classified into four types: single-fidelity, hybrid, adaptive sampling-based, and the MFS models. For the construction of a lightweight digital twin, the MFS models with three features, lightweight, high accuracy, and high robustness, are more suitable. Here, "lightweight" means computationally inexpensive, which takes up less storage space. Among the three features, lightweight is the most important one as it can guarantee the real-time calculation of the digital twin. Besides, both accuracy and robustness are considered. To find an MFS model appropriate to build the lightweight digital twin, several commonly used MFS models have been studied, including co-Kriging [30], MFS-radial basic function (MFS-RBF) [31], MFS-moving least-square (MFS-MLS) [32], and linear regression MFS (LR-MFS) models [33]. The lightweight, accuracy, and robustness of these models were compared with each other comprehensively (see Sec. 3.1).

2.3 Sensor Data. With multiple sensors, the states of physical entities can be dynamically transformed into the digital space, indicating that the sensor data are used as the inputs of the MFS model. Taking these sensor data as dynamic inputs in this way, the structural performance of the system can be predicted in real time once the MFS model is built. The locations where the sensors

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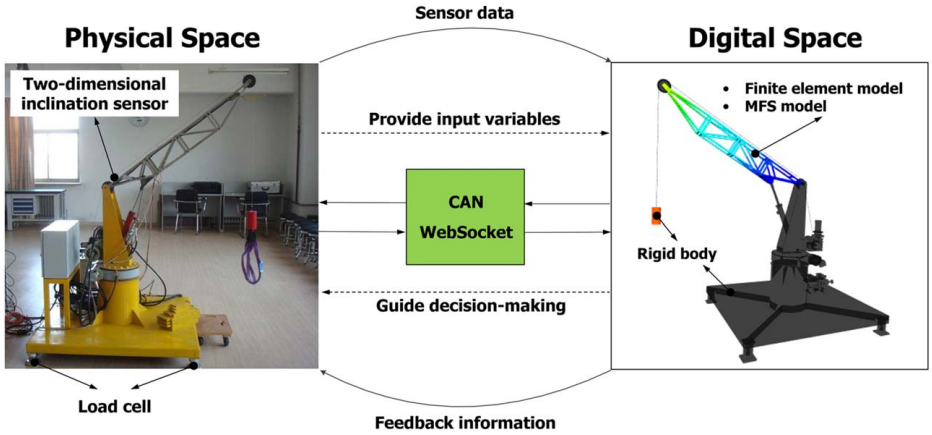


Fig. 1 Framework of the MFS model-based lightweight digital twin for the structural safety monitoring of the crane boom

Table 1 Comparison of the lightweight level among the four MFS models

	MFS-RBF		Co-Kriging		LR-MFS		MFS-MLS	
Properties	Def.	Stress	Def.	Stress	Def.	Stress	Def.	Stress
Occupation of storage space (KB)	36	39	49	46	59	63	48	49
Duration of prediction (s)	0.035	0.036	0.194	0.174	0.116	0.115	2.226	2.266
Duration of training (s)	0.608	0.597	1.635	1.154	0.043	0.042	0.001	0.001

installed are dependent on the inputs of the computer simulation, that is, the task of each sensor is to reflect one dimension of the input variables. For example, the degree of the luffing angle and the mass of loads are two critical variables for the structural analysis of the crane boom. These two variables form an input variable with two dimensions for the computer simulation.

3 Implementation of the Lightweight Digital Twin of the Crane Boom

The framework of the MFS model-based lightweight digital twin for structural safety monitoring is depicted in Fig. 1. In the physical space, the main object is a small-scale boom crane with two types of sensors. The two-dimensional inclination sensor is used to obtain the degree of the luffing angle of the crane boom. The load cells installed on the footplates are employed to check the mass of the lifting load. In the digital space, the combination of the finite element model and the MFS model is adopted to analyze and predict the structural performance of the crane boom. The other components of the boom crane are regarded as the rigid body. To connect the physical space with the digital space, the controller area network and WebSocket are used to build the communication module of the digital twin.

3.1 Selection of Multifidelity Surrogate Models. The duration of prediction, the duration of training, and the occupation of storage space are employed to evaluate the lightweight level of an MFS model used for building the digital twin. Table 1 presents the three properties of four MFS models (mentioned in Sec. 2.2) in which "Def." is the abbreviation of "Deformation." The duration of prediction and training represents the elapsed time during the prediction and training process of each model, respectively. The occupation of storage space means the amount of disk space used when a surrogate model is stored in the JAVASCRIPT object notation format. As shown in Table 1, the occupation of storage space of the MFS-RBF model is less than that of the other three MFS models. The difference in the occupation of storage space will be more obvious when the models become complex. The duration of prediction of the MFS-RBF model is shorter than that of the other three models, which is nearly 1/5 of the co-kriging model, 1/3 of the LR-MFS model, and 1/100 of the MFS-MLS model. The weak aspect of the MFS-RBF model is that it ranks second in terms of the time consumed in the training process among the four models. However, the elapsed time during the training process does not influence the computational efficiency since only the prediction process is performed in the lightweight digital twin. Thus, a model, which completes its prediction process in a short time (<0.05 s), is more suitable for the lightweight digital twin. Therefore, the MFS-RBF model is relatively the most suitable model for the construction of a lightweight digital twin. The detailed building process of MFS-RBF models can be found in our previous work [31].

3.2 Visualization. Figures 2(*a*) and 2(*b*) are the 3D demonstration of the stress and deformation prediction for the crane boom, respectively. As indicated by the black circles 1–4 in Fig. 2(*a*), the main interface of the digital space is divided into four parts: 1 is the LF surrogate model, 2 is the MFS model, 3 is the HF surrogate model, and 4 is the real-time monitoring of the physical crane. The black rectangles 5–7 are the stress/deformation prediction values of the 18 test points. In Fig. 2(*b*), the charts surrounded by black rectangle 8 are used to show data collected from different sensors. The numbers pointed by black circles 9–11 represent the serial numbers of the 18 test points. A video demonstrating the lightweight digital twin of the crane boom can be found online.[2]

3.3 Data Fusion and Information Flow. The data involved in the lightweight digital twin can be divided into three types: geometric data, sensor data, and performance data. The geometric data contain coordinates and indexes of nodes obtained from the HFFE and LFFE models. The sensor data consist of the luffing angle of the crane boom and the mass of loads. The performance data are composed of the high-, low-, and multifidelity stress and

[2]https://youtu.be/H6Hmdt3slqE

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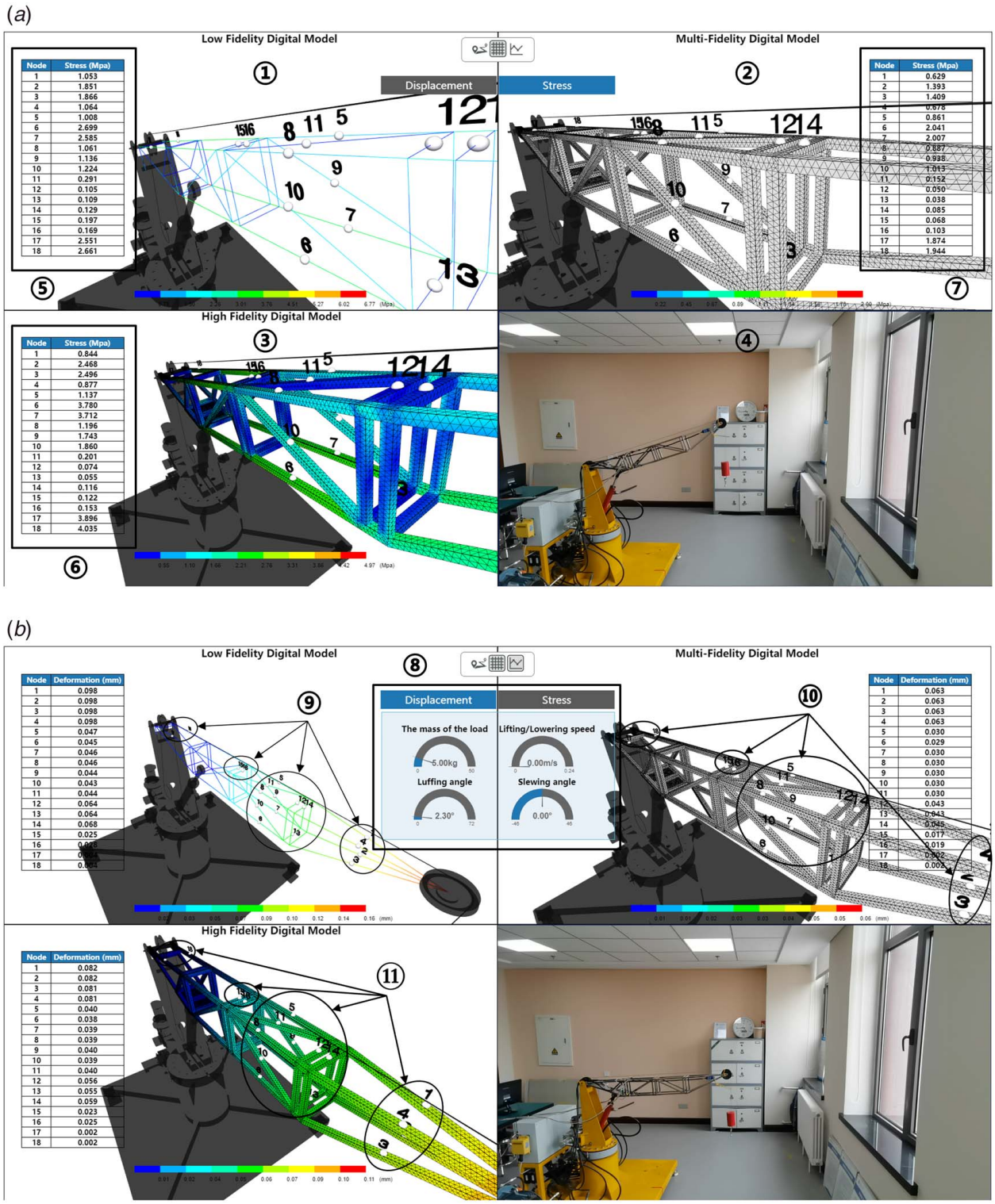


Fig. 2 Visualization of the crane boom rendered in the Web application: (*a*) main interface of the digital space and (*b*) 18 test points in the 3D scene and charts used to show the real-time sensor data

deformation data. In summary, the process to build the lightweight digital twin can be divided into four steps:

- *Step 1:* The geometric data are obtained from a discretization process, where the HFFE and the LFFE solid models are transferred to mesh models. The performance data are obtained by performing the FEM to calculate the deformation and stress on each node of the mesh models.
- *Step 2:* Establish the MFS-RBF model after the performance data is obtained.
- *Step 3:* Collect real-time sensor data as inputs of the MFS-RBF model to dynamically predict the stress and deformation data.
- *Step 4:* Leverage the Web graphics library to combine the geometric and performance data and to render and present the crane boom in a 3D scene.

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Table 2 R^2 provided by evaluating the accuracy of the three models on the 18 test points

	Deformation				Stress			
Test points	HF	LF	MFS	Imp. (%)	HF	LF	MFS	Imp. (%)
1	0.9492	0.8199	0.9882	4.1092	0.8575	0.5626	0.9714	13.2877
2	0.9489	0.8242	0.9882	4.1362	0.9448	0.9448	0.9994	5.7811
3	0.9490	0.8192	0.9882	4.1210	0.9446	0.9472	0.9995	5.8144
4	0.9494	0.8206	0.9882	4.0891	0.8767	0.5571	0.9715	10.8167
5	0.9489	0.8686	0.9882	4.1334	0.9056	0.8453	0.9960	9.9846
6	0.9487	0.8788	0.9881	4.1518	0.9452	0.8037	0.9998	5.7802
7	0.9485	0.8681	0.9881	4.1828	0.9451	0.8433	0.9998	5.7865
8	0.9491	0.8777	0.9881	4.1081	0.9057	0.8100	0.9951	9.8724
9	0.9486	0.9482	0.9881	4.1657	0.9452	0.8956	0.9998	5.7792
10	0.9489	0.9543	0.9881	4.1360	0.9457	0.8866	0.9998	5.7281
11	0.9490	0.9399	0.9882	4.1248	0.9659	−5.1070	0.9899	2.4807
12	0.9490	0.8962	0.9883	4.1427	0.8699	−0.8113	0.9890	13.6847
13	0.9485	0.8924	0.9882	4.1800	0.9462	−8.5465	0.9809	3.6622
14	0.9490	0.8890	0.9883	4.1410	0.9067	0.7074	0.9975	10.0054
15	0.9496	0.9722	0.9881	4.0503	0.9626	−6.9074	0.9887	2.7076
16	0.9495	0.9639	0.9881	4.0671	0.9301	−0.9811	0.9887	6.2948
17	0.9503	−0.7416	0.9869	3.8515	0.9514	0.8960	0.9999	5.0959
18	0.9501	−0.8614	0.9870	3.8909	0.9510	0.8858	0.9999	5.1387

4 Results and Analysis

4.1 Accuracy Verification. The 18 test points shown in Fig. 2(*b*) are used to evaluate the accuracy of three models, including the single-fidelity RBF model trained by HF and LF samples, respectively, and the MFS-RBF model. Table 2 shows that the MFS-RBF model presents a higher accuracy than the single-fidelity RBF models. "Imp." denotes the accuracy improvement of the MFS-RBF model over the RBF model trained by HF samples. It can be seen that most R^2s of the LF RBF model are lower than that of the HF RBF model because the prediction of the LF RBF model is typically not numerically equivalent to the real value but only provides a trend, while the HF RBF model can provide more accurate prediction near the HF samples. Thus, the purpose of the MFS-RBF model is to add the information provided by the LF RBF to the HF RBF to improve the accuracy in the area far away from HF samples.

4.2 Uncertainty Analysis. Note that the MFS-RBF model and FEM used in the lightweight digital twin are set to be deterministic. Thus, only the uncertainty from the physical space (i.e., sensors and other factors) is analyzed. The stress and deformation can be predicted by leveraging the measured data as inputs; thus, the uncertainty in the prediction can be obtained. The uncertainty from the physical space needs to be quantified during the operation process. Thus, we consider evaluating the prior estimate of the sensor data based on the data obtained during the running stage.

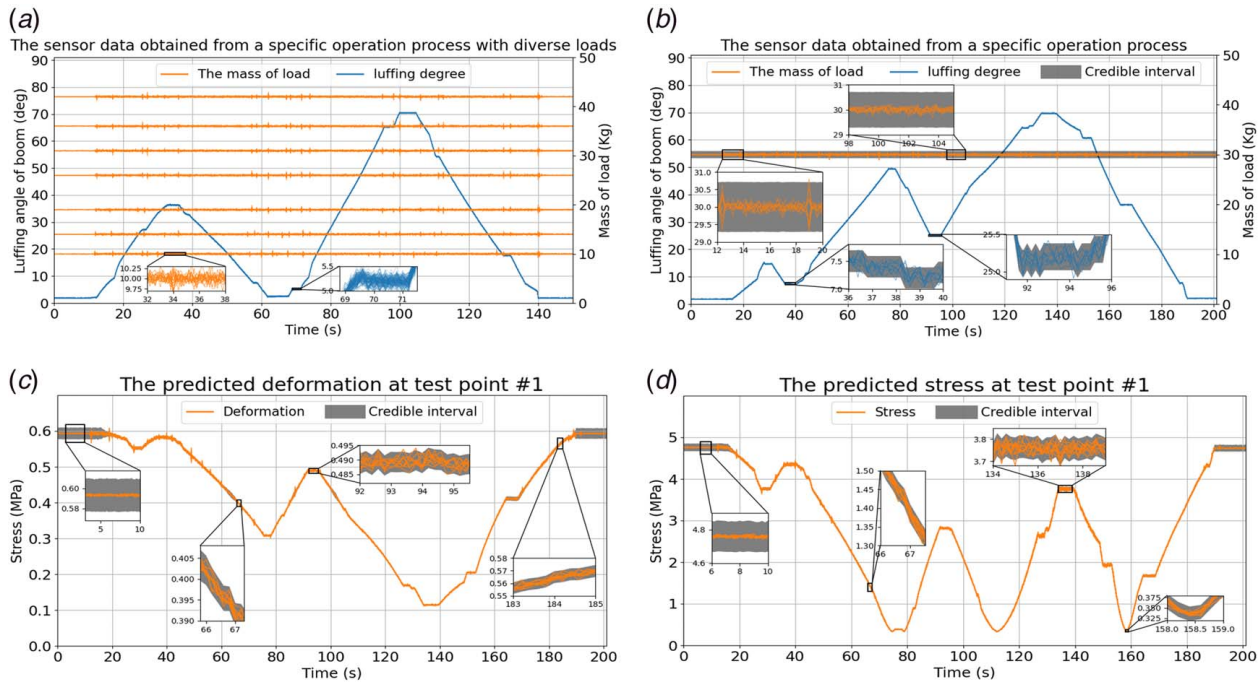


Fig. 3 Uncertainty quantification based on the data collected from the running stage of the crane boom: (*a*) sensor data obtained from a specific operation process, which was repeated 140 times in almost the same trajectory, (*b*) sensor data obtained from another specific operation process, which was repeated ten times in almost the same trajectory, (*d*) deformation prediction at test point #1, (*c*) stress prediction at test point #1

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A specific operation process shown in Fig. 3(*a*) is repeated 140 times in almost the same trajectory during which seven loads of different masses are lifted, and each type of load is lifted 20 times repeatedly. Correspondingly, 140 groups of stress and displacement data are generated, respectively. At each sampling point, we model the uncertainty using the Gaussian distribution and estimate the parameters in the distribution by the sensor data and predicted value (i.e., stress or deformation). Consequently, two RBF models are built to obtain the relationship between means and standard deviations of the load and the luffing angle, respectively. In addition, two RBF models are used to establish the relationship between the means of sensors (a two-dimensional data composed of load and luffing angle) and the standard deviations of the stress and the deformation, respectively. The 95% credible intervals (light gray band) shown in Figs. 3(*b*)–3(*d*) achieve good uncertainty quantification in both the static and operating stages of the boom. By considering the unknown uncertainty caused by the operation, the credible interval can be adaptable so that a more reasonable estimation of the uncertainty can be provided.

5 Discussion and Conclusions

This article presents an efficient way to build a lightweight digital twin of a crane boom for structural safety monitoring based on the MFS model. Different from the previous method, which only uses numerical methods or single-fidelity surrogate models to build digital twins, the MFS model-based lightweight digital twin combines their advantages while eliminating their shortcomings. Taking the MFS-RBF model as an example, we proved that the proposed digital twin can improve the accuracy of digital twins built by single-fidelity surrogate models and reduce the computational costs of digital twins built by numerical methods. In addition, the uncertainty from the physical space (i.e., sensors and other factors) was analyzed to quantify the uncertainty of the lightweight digital twin. The effectiveness of the proposed digital twin was validated by applying it on a lab crane boom. In the MFS model-based digital twin, the single-fidelity LF surrogate model trained by considerable LF data are approximately regarded as the mechanistic model over the entire solution space, while the single-fidelity HF surrogate model built by a small amount of HF data is employed as the statistical model. Therefore, the MFS model can integrate the advantages of the mechanistic model that can provide a causal relationship in structural analysis and the characteristics of statistical models that are easy to build and less costly. Thus, the MFS model-based lightweight digital twin is reliable and computationally affordable, enabling real-time structural safety monitoring. Moreover, the MFS model-based lightweight digital twin is not only suitable for mechanical structures, equipment, and systems but also can be extended to other fields and industries, such as biomechanics and fluid mechanics.

This work was limited to predicting the structural performance of a linear elastic material, and the structural analysis is only restricted to the elastic stage. Future work should, therefore, be designed to explore more complex physical processes, such as nonlinear materials, plastic deformation, and multiphysics coupling because a more complex model can more closely reflect complex physical processes in real life and can provide a more realistic digital twin.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request. The data and information that support the findings of this article are freely available at: sxg@dlut.edu.cn.

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